**BizRadar v4**

# Universal Sector Algorithm

Perfect-fit specifications for every NAICS sector — built so a single platform can analyze any of 1,464 business types with the same rigor it applies to a hotel.

| | | | |
|---|---|---|---|
| 1,464 business types | 20 NAICS sectors (18 in scope) | 63 sub-cluster specs | 5 implementation phases |

### Why this document exists

BizRadar v3 is a hospitality intelligence engine pretending to be universal. It works beautifully for hotels, restaurants, salons, gyms — and falls apart for the other ~67% of businesses on the master list. v4 introduces a Master Router (Layer 0.5) that classifies every input by business model and routes it to a sector-specific pipeline. The pipeline architecture stays the same; the contents of every layer are swapped per sector. This document is the complete specification of those contents — one chapter per sector, plus the universal scaffolding and the implementation phases.

### How to read this

Part 1 is the architecture and the router — read top to bottom. Part 2 has one chapter per sector — read the chapters relevant to your phase. Part 3 is the failure-rate database. Part 4 is the implementation roadmap with explicit phase gates. Part 5 is reference and appendices. Engineers should pair this document with the existing v3 server.js — every section maps to a code module that needs to be added or modified.

### Promise of this spec

Every sector chapter is structured identically: business-model classification, sub-clusters, data sources, review taxonomy, opportunity playbook, specificity rules, cost scale, persona schema, cycle model, issue library, report sections, KPIs, and out-of-scope notes. If a section says 'not applicable' for a sector, that is a deliberate engineering decision — not an oversight. The router is responsible for routing requests to the correct chapter; the chapter is responsible for producing a useful report.

# Contents

## Part 4 — Implementation Roadmap



## Part 5 — Appendices

**PART 1 — UNIVERSAL ARCHITECTURE**

# 1.1 The flaw of v3 in one diagram

BizRadar v3 has one pipeline. Every business — a hotel, a CPA firm, a steel forge, a SaaS company, a hog farm — runs through the same nine layers with the same hospitality-shaped contents inside. The architecture is sound; the contents are wrong for everything except hospitality.

| | | | |
|---|---|---|---|
| L0 Input | Classify input type | Business name regex includes hotel chains | Trucking firm or CPA firm gets misrouted |
| L1 Data | 9 parallel APIs | Google + TripAdvisor + Foursquare | B2B firms have ~0 reviews on these |
| L2A Reviews | Synthesize reviews | Hotels have hundreds of reviews | Manufacturer has 3 — analysis is noise |
| L2A Categories | Tag review categories | food, ambiance, cleanliness, WiFi | Wrong taxonomy for a dental practice |
| L2A Competitors | Top 3 within 2.5km | Walk-in radius | B2B competes statewide or nationally |
| L3 Opportunities | 18 categories of ideas | Sensory experience, photo moments, scents | Meaningless for a wholesaler or a SaaS |
| L4 Hyper-local | Named producers within 60mi | Carr Valley Cheese style local | B2B 'local' is industries, not producers |
| L5 Scoring | Cost scale 1–5 = $0 to $10k | Hotel-scale spending | Manufacturer evaluates in $100k–$10M |
| L6 Personas | Sarah, 45, parent for graduation | B2C consumer schema | B2B buyer has committee, not lifestyle |
| L7 Seasonal | Calendar quarters | Tourist seasons | Construction, B2B run on different cycles |
| L8 Issues | WiFi, bedbugs, noise | Hospitality complaint library | HVAC complaints are no-shows + surprise pricing |
| L9 Output | 9 fixed sections | Hospitality report layout | B2B needs RFP pipeline + cert roadmap |

**The fix in one sentence**

Keep the 9-layer architecture. Insert a Master Router at Layer 0.5 that selects which **contents** populate each subsequent layer based on the business model class. The architecture is universal. The contents are sector-specific. This document specifies both.

## 1.2 The v4 architecture — router plus 9 layers

v4 inserts Layer 0.5 — the Master Router — between input classification (L0) and data collection (L1). The router reads the business model class from L0, then selects the appropriate sector profile. Every downstream layer then loads its contents from the selected profile.

```
USER INPUT
    |
    v
L0   Input Classifier  ->  detect: location | city | niche | url | ambiguous
    |
    v
L0.5 MASTER ROUTER  (NEW IN v4)
       reads: name + Google Places category + NAICS keyword + Part2 lookup
       outputs: { naics_code, business_model, sector_profile_id }
       loads:   sector_profiles[profile_id]  ->  used by every layer below
    |
    v
L1   Parallel Data Collection
       APIs and competitor radius come from the loaded profile
       e.g. hotel = Google + TA + Yelp / 2.5km
            HVAC  = Google + Angi + BBB / 30 mi
            SaaS  = G2 + Capterra + ProductHunt / global
    |
    v
L2A  Reviews & Competitor Analysis
       review categories come from profile
       commonality detection unchanged
    |
    v
L3   Opportunity Engine
       18 categories <-- HOSPITALITY playbook
       12-20 categories <-- per-sector playbook (one of 18 sets)
    |
    v
L4   Specificity Layer (was 'Hyper-Local' in v3)
       what counts as a real named entity comes from profile
    |
    v
L5   Scoring & Ranking
       cost scale relative to revenue band, set by profile
    |
    v
L6   Persona Engine
       schema picked by profile: B2C-consumer | B2B-buyer | patient | client
    |
    v
L7   Cycle Model (was 'Seasonal' in v3)
```

```
       cycle definition picked by profile
    |
    v
L8   Issue Library
       per-sector category-standard complaints loaded from profile
    |
    v
L9   Output Delivery
       section list picked by profile (some appear / disappear by sector)
    |
    v
REPORT DELIVERED TO USER
```

**What this means in code terms**

v4 introduces a new file: **sector_profiles.js**. It exports a dictionary keyed by **sector_profile_id**. Each value is a complete profile object with fields for: data sources, review categories, opportunity playbook, specificity rules, cost scale, persona schema, cycle model, issue library, and report sections. The Master Router selects one profile per request. server.js is refactored to read from the loaded profile rather than from hardcoded hospitality logic.

# 1.3 Layer 0.5 — Master Router (full spec)

The Master Router runs after L0 and before L1. It outputs three values that determine the entire downstream pipeline: **naics_code** (6-digit), **business_model** (one of nine classes), and **sector_profile_id** (the lookup key into sector_profiles).

## Inputs to the router

- **Business name** — from L0 (e.g. 'Smith Plumbing', 'AmericInn Dodgeville')
- **Google Places category** — from L1 once it returns (router can run after L1 for ambiguous cases)
- **NAICS keyword library** — appendix E in this document, mapping ~3,000 keywords to NAICS codes
- **Part 2 master list** — 452 modern business types that map to NAICS parents
- **User override (optional)** — explicit NAICS code if user is sophisticated

## Classification rules — order matters

1. Try exact match against Part 2 master list (e.g. 'food truck' -> 722330, 'yoga studio' -> 713940).
2. Try Google Places primary category mapping (e.g. 'lodging' -> 721110, 'plumber' -> 238220).
3. Try keyword regex from Appendix E (e.g. 'CPA' / 'tax' / 'accountant' -> 541211).
4. If still ambiguous, surface a confirmation question to the user before any L1 calls fire.
5. Record confidence score for the classification — under 0.6 triggers a soft-confirm dialog.

## The nine business model classes

| | | | |
|---|---|---|---|
| B2C_walk_in | Customer physically walks in, mostly without appointment | Foot traffic, on-premise consumption | 722511, 445110, 812111 |
| B2C_appointment | Consumer appointments, sometimes drop-in | Scheduling system, professional license | 621210, 541110, 812112 |
| B2C_online | Direct-to-consumer online only | No physical address that matters | 454110, 519290, 513210 |
| B2B_local | Sells to local businesses, geography matters | Service area, route density | 238220, 561720, 484110 |
| B2B_regional | Sells to businesses in a multi-state region | Distributor / wholesale model | 423110, 484121, 561311 |
| B2B_national | National or global B2B sales | Trade show circuit, named OEMs | 33xxxx, 5415xx, 423430 |
| B2G | Sells primarily to government entities | SAM.gov registration, set-aside certs | 237310, 541330, 561612 |
| Regulated_monopoly | Geographic monopoly under regulator | PUC / FERC oversight | 221xxx, 4861xx, 4862xx |

| | | | |
|---|---|---|---|
| Out_of_scope | Public administration, holding companies | Not a customer-facing business | 5511xx, 9211xx-9281xx |

## Sector profile selection

Once class is determined, the router picks one of 60+ sector profile IDs. The profile ID is more granular than the NAICS sector — many sectors have multiple profiles (e.g. construction has separate profiles for residential trades, commercial GCs, and civil/heavy). The full mapping is in Appendix A.

## Pseudocode

```
function classify(input) {
  // Step 1: extract candidate NAICS
  let candidates = [];
  candidates.push(...part2_lookup(input.name));        // food truck -> 722330
  candidates.push(...gplaces_to_naics(input.gcategory));// 'plumber' -> 238220
  candidates.push(...keyword_regex(input.name));        // 'CPA' -> 541211
  if (candidates.length === 0) return ASK_USER;

  // Step 2: pick highest-confidence candidate
  let { naics, confidence } = best(candidates);
  if (confidence < 0.60) return SOFT_CONFIRM(naics);

  // Step 3: derive business model + profile
  let business_model = naics_to_model[naics.slice(0,4)];
  let profile_id     = naics_to_profile[naics.slice(0,4)];

  // Step 4: return full classification
  return { naics, business_model, profile_id, confidence };
}
```

## Failure modes the router must handle

| | | |
|---|---|---|
| Ambiguous name | 'Smith & Sons' could be 50 things | Soft-confirm dialog — show top 3 NAICS guesses |
| Out-of-scope sector | User pastes a government office | Friendly rejection with explanation, refund flow |
| Multi-line-of-business | Hotel with restaurant inside | Pick primary NAICS, note secondary in report |
| Brand new format | AI-driven business with no NAICS yet | Map to closest parent + add to Part 2 backlog |
| User override mismatch | User claims NAICS that conflicts with Google | Use user override but flag the mismatch in QA log |

# 1.4 Universal scaffolding — what stays constant across all sectors

These elements do not change between profiles. They are the bones that hold the system together. Anything not on this list *must* be sector-customized.

## Constants across every sector

| | | |
|---|---|---|
| Layer execution order | L0 -> L0.5 -> L1 -> L2A -> L3 -> L4 -> L5 -> L6 -> L7 -> L8 -> L9 | Section 1.2 |
| Promise.all() data collection | Parallelism is the right pattern regardless of which APIs are called | L1 |
| 8-second hard timeout per source | Fail fast principle is universal | L1 |
| Recency weighting on review scores | More recent feedback is more relevant in every sector | L2A |
| KNOWN vs HIDDEN issue classification | Known to all competitors -> capped. Unique -> high priority | L8 |
| Final scoring formula shape | (Impact x 2 + Novelty x 1.5 + Mention x 1) / (Ease + Cost) | L5 |
| Quick Win definition (relative) | High impact + low cost + low ease — content-relative thresholds | L5 |
| Specificity rule (zero-tolerance) | Every recommendation names a real entity — what kind varies | L4 |
| Confidence score on output | Reports always include data quality + recommendation confidence | L9 |
| Cache 24h on identical input | Cost / latency optimization is universal | L1 |

## Quality gates that fire for every report

- **Specificity gate:** if any recommendation contains a banned generic phrase ('local farm', 'nearby business', 'community partner'), force re-generation.
- **Volume gate:** minimum 12 ranked recommendations, minimum 3 personas, minimum 3 cycle entries — pad does not count.
- **Evidence gate:** every issue must have at least one quote (or one structured datum for B2B) under 15 words.
- **Confidence gate:** if data quality average is below 0.4, prepend a yellow warning to the report.
- **Out-of-scope gate:** if business_model is 'Out_of_scope', do not generate — return refund-eligible message.
- **Hallucination gate:** every named entity (producer, partner, certification) must be verifiable in a structured source — not free-text Claude output.

**What "sector-perfect" means operationally**

A report is sector-perfect when (a) every layer's contents come from the right profile, (b) every quality gate passes, (c) the user reads three recommendations and says "I haven't heard any of this before — and it's true". Generic, hospitality-shaped, or wrong-radius output fails this test and ships zero value to a non-hospitality customer.

# 1.5 The substitution principle — what changes per sector

Each sector profile is a JSON-serializable object with the following ten fields. Every chapter in Part 2 fills in these ten fields for one or more sub-clusters of a sector.

| | | | |
|---|---|---|---|
| 1 | data_sources | list[Source] | Which APIs L1 fires, and the fallback chain |
| 2 | competitor_radius | string \| mi \| rule | How L1 defines 'nearby competitor' |
| 3 | review_categories | list[string] | What L2A tags reviews with |
| 4 | opportunity_playbook | list[Category] | The 12-20 categories L3 researches |
| 5 | specificity_rules | list[Rule] | What counts as a real named entity in L4 |
| 6 | cost_scale | ScaleSpec | Currency bands for L5 'Cost' dimension |
| 7 | persona_schema | SchemaName | Field set for L6 personas |
| 8 | cycle_model | CycleSpec | L7 cycle definition (calendar / fiscal / event-driven) |
| 9 | issue_library | list[KnownIssue] | L8 category-standard complaints with quote templates |
| 10 | report_sections | list[SectionId] | L9 section list, in the order they render |

## Profile object shape (TypeScript-ish)

```
interface SectorProfile {
  id: string;                          // e.g. 'hospitality.hotel'
  display_name: string;                // 'Hotels & Motels'
  naics_codes_covered: string[];       // ['721110','721120','721191',...]
  business_model: BusinessModel;
  data_sources: Source[];              // ordered, with fallback chain
  competitor_radius: RadiusRule;       // miles | service_area | national
  review_categories: string[];         // 6-12 tags for L2A
  opportunity_playbook: Category[];    // 12-20 entries
  specificity_rules: Rule[];           // what L4 must name
  cost_scale: { unit: 'USD', bands: [number,number,number,number] };
  persona_schema: 'B2C' | 'B2B' | 'patient' | 'client';
  cycle_model: { kind: 'calendar' | 'fiscal' | 'event'; entries: Cycle[] };
  issue_library: KnownIssue[];
  report_sections: SectionId[];        // ordered list of L9 sections
  out_of_scope_naics: string[];        // 6-digit NAICS to refuse even if matched
}
```

*Every sector chapter in Part 2 is essentially a verbose, human-readable filling-in of this object. An engineer can read a chapter and translate it directly to the JSON profile.*

# 1.6 Quality gates — universal pre-flight checks

These gates run after L9 builds the report but before it is shown to the user. Any gate that fails either triggers re-generation (soft fail) or blocks the report and refunds the user (hard fail).

| | | | |
|---|---|---|---|
| Specificity | Zero generic phrases (banned word list) | Soft | Re-prompt L3 / L4 with stricter instructions |
| Volume | Min 12 recs, 3 personas, 3 cycle entries | Soft | Re-prompt with minimum count enforcement |
| Evidence | Every issue has a quote / data source | Soft | Drop unsupported issues, re-prompt for replacements |
| Confidence | Avg data quality > 0.4 | Hard | Prepend warning banner; user can still view |
| Out of scope | business_model not in {Out_of_scope} | Hard | Refund + explain; do not generate |
| Hallucination | Every named entity is in structured source | Hard | Strip unverified entities, re-prompt to fill gaps |
| Coverage | All 10 profile fields populated | Hard | Engineering alert — profile is incomplete |
| Cost sanity | No recommendation outside cost band | Soft | Re-rank with cost band enforced |
| Persona match | All personas use the profile's schema | Soft | Re-prompt L6 with schema reference |
| Section presence | All required sections non-empty | Soft | Either fill or remove from report |

**Why hallucination is a hard fail**

If BizRadar tells a Wisconsin hotel to partner with 'Driftless Cheese Co.' and that company does not exist, the user catches it on the first phone call and never trusts the product again. v3 has caught this exact failure mode three times in testing. v4 enforces structured-source verification: every named entity must be present in Google Places, Census Business Patterns, USDA producer registry, ThomasNet, or another verifiable source. Free-text Claude output without source linkage is rejected.

**BUILD STATUS — THIS IS A PARTIAL DOCUMENT**

# What is in this version

This is the first batch of the BizRadar v4 spec. Front matter (Part 1) is complete. Three of the eighteen in-scope sector chapters are complete (Agriculture / Mining / Utilities — the heavy-industrial trio). The remaining fifteen sectors are queued and will be appended in subsequent batches. Parts 3 (failure-rate database), 4 (implementation roadmap), and 5 (appendices) are also queued.

## What is complete

| | |
|---|---|
| Part 1 — Universal Architecture | Done — 6 sections covering router, scaffolding, substitution principle, quality gates |
| Ch. 11 — Agriculture, Forestry, Fishing | Done — 3 sub-clusters: commodity / agritourism / forestry |
| Ch. 21 — Mining, Oil & Gas | Done — 2 sub-clusters: aggregates / oil-gas-metals |
| Ch. 22 — Utilities | Done — 1 sub-cluster: electric/gas/water utility |

## What is queued for the next batches

| | |
|---|---|
| Ch. 23 — Construction | Phase 2 priority — residential trades, commercial GCs, civil/heavy |
| Ch. 31-33 — Manufacturing | Phase 4 — food/bev DTC, industrial B2B, advanced (346 codes — biggest sector) |
| Ch. 42 — Wholesale Trade | Phase 4 — distributors and rep firms |
| Ch. 44-45 — Retail Trade | Phase 1 — closest fit to v3, brick-and-mortar + auto + e-commerce |
| Ch. 48-49 — Transportation/Warehousing | Phase 4 — trucking, transit, courier, warehouse |
| Ch. 51 — Information | Phase 5 — SaaS, media, telecom, publishing |
| Ch. 52 — Finance & Insurance | Phase 3 — banking, insurance agencies, RIAs |
| Ch. 53 — Real Estate & Rental | Phase 3 — agents, property management, self-storage |
| Ch. 54 — Professional Services | Phase 3 — legal, accounting, A&E;, consulting, vet, photography |
| Ch. 56 — Admin & Support / Waste | Phase 4 — staffing, janitorial, pest, security, waste |
| Ch. 61 — Educational Services | Phase 5 — tutoring, K-12, bootcamps, trade schools |

| | |
|---|---|
| Ch. 62 — Healthcare & Social Assistance | Phase 1 — medical, dental, mental health, child care |
| Ch. 71 — Arts/Entertainment/Recreation | Phase 1 — mostly fits v3 with sub-cluster tweaks |
| Ch. 72 — Accommodation & Food Services | Phase 0 — port v3 logic into profile format (the sweet spot) |
| Ch. 81 — Other Services | Phase 1/2 — auto repair, personal care, funeral, laundry, civic |
| Ch. 55 + 92 — Out of scope | Friendly rejection logic for holding co + public admin |
| Part 3 — NAICS-4 Failure-Rate Database | 300+ codes with year-1/3/5/10 survival data |
| Part 4 — Implementation Roadmap | 5 phases with explicit gates and engineering tasks |
| Part 5 — Appendices | NAICS map, data sources, code structure, glossary, classifier keywords |

**How to continue the build**

Reply with 'continue' and the next batch of sector chapters will be appended. Recommended batch order: (1) Hospitality + Retail + Recreation [Phase 0/1, easiest], (2) Construction + Healthcare + Other Services [Phase 1/2, B2C appointment + trades], (3) Professional Services + Real Estate + Finance [Phase 3], (4) Manufacturing + Wholesale + Transportation + Admin [Phase 4 B2B], (5) Information + Education + Out-of-scope + Parts 3/4/5 [Phase 5 + assembly].

**PART 2 — SECTOR CHAPTERS**

# Per-sector specifications

Each chapter below is structured identically: business-model classification, sub-clusters, data sources, review taxonomy, opportunity playbook, specificity rules, cost scale, persona schema, cycle model, issue library, report sections, KPIs. Engineers should treat each chapter as a fillable template for the sector_profiles.js file. Anything marked 'not applicable' is a deliberate engineering decision — not an oversight.

**Reading order recommendation**

Read the chapter that maps to your current implementation phase. Phase 1 starts with Hospitality (Ch. 72) as the v3 sweet spot, then Healthcare (Ch. 62) and Personal Care (in Ch. 81) as the highest-payoff B2C-appointment additions. Phase 2 moves to Construction trades (Ch. 23). Phase 3 tackles Professional Services (Ch. 54). Phase 4 handles B2B (Ch. 31-33, 42, 51). Phase 5 covers specialty branches (Agriculture, Mining, Utilities, Real Estate, Education).

## CHAPTER 11 — AGRICULTURE, FORESTRY, FISHING, HUNTING

Commodity producers, ag-tourism operators, and forestry / fishing operations. The hospitality pipeline does not work because most of these have zero Google reviews and sell to processors, not consumers.

*Covers 64 NAICS codes from Part 1 of the master list, plus modern formats from Part 2.*

**The v3 problem in this sector**

A 1,200-acre soybean operation has no Google reviews, no walk-in customers, and no nearby competitors in the hospitality sense. v3 produces an empty report. v4 splits this sector by output channel: commodity producers vs direct-to-consumer ag operations.

### Sub-clusters in this sector

This sector splits into 3 sub-clusters because the business models inside diverge enough that one profile cannot serve them all. Each sub-cluster gets its own complete spec below.

| | | | |
|---|---|---|---|
| agriculture.commodity | Commodity producers (crops, livestock, dairy, aquaculture) | B2B_regional | 111110, 111150, 112111, 112120, 112511 — and the rest of 111xxx / 112xxx |
| agriculture.agritourism | Agritourism, U-pick, farm stands, agritainment | B2C_walk_in | 111219, 111421, 111998, 112910 (apiculture for honey DTC) — usually rolled up to parent farming code |
| agriculture.forestry_fishing | Forestry, logging, commercial fishing, support services | B2B_regional | 113110, 113210, 113310, 114111, 114112, 114119, 114210 |

# 11.commodity — Commodity producers (crops, livestock, dairy, aquaculture)

**Business model:** B2B_regional | **Profile ID:** agriculture.commodity

**Covers NAICS:** 111110, 111150, 112111, 112120, 112511 — and the rest of 111xxx / 112xxx

**Modern formats (Part 2):** Mushroom Farm (Specialty), Microgreens Producer, Hydroponics, Hemp Farm

### 1. Data sources (replaces v3 hospitality stack)

| | | | |
|---|---|---|---|
| USDA NASS QuickStats | County yields, prices received, planted acres | Free, public | USDA AgCensus 5-yr |
| USDA AMS market reports | Daily commodity prices, futures basis | Free | CME settlements |
| USDA FSA office directory | Local FSA agent + named programs (CSP, EQIP, CRP) | Free | Extension office |
| Land grant extension service | Named extension agent + variety trial results | Free | University ag dept |
| Farm Service Agency loan officer | Named local FSA loan officer | Free | Farm Credit branch |
| Ground-truth grain elevator list | Named local elevators + cash bids | Scrape / API | Direct call |
| LandWatch / AcreValue | Comparable land values + rental rates | Subscription | County assessor |
| NOAA / USDA RMA crop weather | Heat units, rainfall vs normal, drought monitor | Free | NWS station |

## 2. Competitor radius

Three radii: (1) within 25 miles for shared elevators/processors and labor pool, (2) within 100 miles for buyer competition, (3) state-wide for commodity-class peers. v3's 2.5km is meaningless here.

## 3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Yield vs county average — quantitative, not review-based
- Soil health trajectory — soil tests over time
- Crop quality / grade — protein, falling number, test weight
- Labor reliability — H-2A retention, local crew availability
- Equipment uptime — downtime hours per season
- Buyer relationships — repeat business with named processors
- Insurance / risk profile — RMA loss ratio over 5 years
- Conservation participation — CSP / EQIP enrollment status

## 4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **18** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

| | | |
|---|---|---|
| Specialty / heirloom premium varietals | Named varietals with documented price premiums in this region | Switch 200 acres of #2 yellow corn to Floriani Red Flint — sells to Bob's Red Mill at $0.42/lb premium vs $0.18/lb commodity |

| | | |
|---|---|---|
| Organic / Real Organic / Regenerative cert | Cert body + transition timeline + premium captured | Real Organic Project certification — 12-month transition — $0.85/bu premium on soybeans into Patagonia Provisions supply chain |
| Direct-to-processor named contract | Named processors within 200 miles seeking your commodity | Anheuser-Busch Wisconsin Rapids barley contract — locked basis, $0.50/bu over CBOT |
| DTC / farm-stand pivot for a portion | Highway frontage, traffic counts, complementary local destinations | 5% of acreage to pumpkins / sweet corn for $40k stand revenue Oct-Dec on US-151 frontage |
| CSA / share program | County household income + organic demand + delivery zones | 60-share CSA — Madison drop point — $700/share — $42k revenue from 2 acres |
| Regen ag certification + carbon credit | Indigo, Nori, Truterra, Bayer ForGround pricing | Truterra Sustainability Insetting — $20/acre baseline plus stacking with Climate-Smart commodity |
| Conservation easement income | NRCS Wetland Reserve / state ag easement programs | WI Knowles-Nelson stewardship easement — $2,400/acre on 80 marginal acres = $192k one-time |
| Co-op participation upgrade | Named regional co-op + member benefits unused | CHS Inc patronage — increase share purchase to qualify for top-tier dividend rate |
| Variety trial partnership | Land-grant university researcher + on-farm trial program | UW-Madison soybean variety trial — $2k stipend + free seed + first access to released line |
| Custom farming for a neighbor | Aging farmer within 10 miles + idle equipment hours | Custom-farm 600 acres for Smith Bros — $80/ac corn, $65/ac beans = $87k margin on idle equipment |
| Equipment-share co-op | Named neighbors with overlapping but non-conflicting needs | 4-farm machinery co-op for 16-row planter — split $480k equipment $120k each vs solo |
| Cover crop seed sales | Local seed dealer relationships + cover crop adoption rates | Add cereal rye production for sale to Albert Lea Seed — $12/bu vs commodity $4.50 |
| Custom hay / forage premium | Named horse barns / dairies needing high-RFV hay | 180-RFV alfalfa to named horse boarding barn — $375/ton vs commodity $180/ton |
| On-farm storage premium capture | Storage capacity vs production + basis seasonal pattern | Add 50,000 bu bin — capture $0.40/bu post-harvest basis recovery = $20k/yr at 6% borrowing cost |
| Trucking / logistics in-house | Per-mile cost vs paid hauling + return trip backhauls | Buy used semi for $45k — eliminate $0.18/bu hauling on 90k bu = $16k/yr after fuel + repairs |

| | | |
|---|---|---|
| Solar lease income | Utility solar developers + capacity zone in interconnect queue | Alliant Energy 80-acre solar lease — $1,200/ac/yr vs $185/ac/yr cash rent on marginal land |
| Hunting / recreational lease | Named outfitters + waterfowl flyway / deer harvest density | Pheasants Forever guided hunt lease — 320 acres CRP — $3,500/season + 6-day hunting rights |
| Succession / transition plan | Named ag attorneys + trust structures + Section 2032A | Iowa State CSBE estate plan — qualified family farm — $4.5M estate tax savings via 2032A |

## 5. Specificity rules (L4 — what 'real named entity' means here)

- Every named buyer must be a real grain elevator, processor, or co-op verifiable in USDA listings or state ag department
- Every named program must be a real USDA, NRCS, or state program with a current sign-up window
- Every variety / cultivar must be a real released line with documented yield data
- Every named extension agent / FSA officer must be a real person currently in that role
- Every named buyer contract must reference current published basis or the commodity's price discovery point (CBOT, KCBT, etc.)

## 6. Cost scale (L5 — per-sector currency bands)

Farm operations spend across a wider range than hospitality. The cost scale is calibrated to typical farm operating budgets:

| | |
|---|---|
| 1 — Free or under $500 | Sign up for a program, change a marketing decision, switch a buyer |
| 2 — $500 to $5,000 | Soil tests, cover crop seed, fence repair, small equipment add-on |
| 3 — $5,000 to $50,000 | Variety switch on portion of acres, on-farm storage retrofit, cert transition |
| 4 — $50,000 to $250,000 | New grain bin, used semi, small DTC building, organic transition |
| 5 — Over $250,000 | Major equipment, conservation easement (positive cashflow), land purchase |

## 7. Persona schema (L6)

**Schema:** B2B_buyer (the customer is a procurement manager at a processor / elevator / co-op, not a consumer)

Required persona fields:

- Buyer name + role + company (e.g. "Tom Reynolds, grain merchandiser, Cargill Soy Crush, Cedar Rapids")
- Volume needed annually + delivery window
- Quality specs that drive premiums (protein, oil, FFA, mycotoxin)
- Decision criteria order (price, reliability, quality, location, relationship)
- Contract structure preferred (forward, basis, no-price-established)

- Pain points the buyer experiences with current suppliers
- How this farm differentiates vs the buyer's other suppliers

| **Example persona: Tom Reynolds, grain merchandiser, Cargill Soy Crush — Cedar Rapids** |
|---|
| Buys 4M bushels of #1 soy/yr within 90mi. Pays $0.10/bu premium for protein > 35.5% delivered Aug-Oct. Pain point: small farms don't test protein, so he assumes commodity grade. Differentiation: send protein test results with each load — qualifies for the premium tier automatically. LTV at this farm: 60,000 bu/yr × $0.10 = $6,000/yr in premiums alone. |

## 8. Cycle model (L7 — replaces calendar seasons)

Farm cycles are driven by crop calendar, not tourist season. Each entry below maps a window to specific marketing / operational actions.

| | | |
|---|---|---|
| Pre-plant (Feb-Apr) | Seed orders + crop insurance deadline | Lock variety contracts, finalize FSA acreage, complete RMA elections |
| Plant + early season (Apr-Jun) | Cash flow trough | Operating loan draws, custom planter scheduling, in-season N decisions |
| Mid-season (Jun-Aug) | Crop scouting + side-dress + yield projection | Forward sell 30-50% of expected production at favorable basis windows |
| Pre-harvest (Aug-Sep) | Combine prep + storage + trucking | Lock harvest hauling rates, secure custom combine if needed, line up labor |
| Harvest (Sep-Nov) | Peak revenue + peak labor + peak risk | Direct-ship contracted bushels, hold uncontracted in on-farm storage for basis recovery |
| Post-harvest marketing (Nov-Mar) | Basis recovery + tax-year decisions | Sell stored grain on basis recovery, prepay inputs for next year, fund retirement plans |
| Off-season agritourism (Oct-Dec, Apr-May) | Pumpkin / Christmas tree / U-pick windows | Peak DTC revenue if a portion of operation is direct-to-consumer |

## 9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Weather risk dominates outcomes — mention but never lead the report with it
- Commodity price exposure — universal — mention via marketing plan, not as a "fix"
- Labor shortage — universal across H-2A users — frame as opportunity for automation / housing
- Succession / transition stress — universal in farms aged 55+ — frame as planning opportunity
- Equipment downtime during critical windows — universal — frame as backup-machinery / co-op
- Buyer concentration risk — universal — frame as diversification opportunity
- Land rent inflation outpacing yield gains — universal — frame as conservation / specialty pivot
- Carbon / nitrate regulatory pressure — emerging — frame as cover crop / 4R nutrient planning

## 10. Report sections (L9 — section list and order)

1. **Operation snapshot** — Acres, crops, livestock head, named buyers, 5-yr yield trend vs county
2. **Top 5 unique opportunities** — Drawn from the 18 categories above — at least 3 must be premium-capture, not cost-cut
3. **Buyer / processor map** — Named buyers within 200mi + unused contracts + premium opportunities
4. **Program / grant pipeline** — Named USDA + state programs with deadlines and dollar amounts
5. **Cycle action plan** — 7-window calendar with specific actions per window
6. **Risk register** — Named risks with mitigation, ranked by farm-level $ exposure
7. **Succession / transition flag** — If operator > 55, surface a "talk to an ag estate attorney" line item
8. **Known industry challenges** — Weather / price / labor / regs — collapsed bottom drawer

## Key performance indicators worth tracking

- Yield index vs 5-year county average (Bushels/ac actual ÷ county avg)
- Premium captured per bushel ($/bu over commodity board price)
- Direct-to-consumer revenue as % of total (target: 5-15% to diversify)
- Conservation / program income as % of total (target: 5-20%)
- Operating expense ratio (operating cost ÷ gross revenue, target < 0.75)
- Working capital months (current assets ÷ monthly burn, target > 6 mo)

# 11.agritourism — Agritourism, U-pick, farm stands, agritainment

**Business model:** B2C_walk_in | **Profile ID:** agriculture.agritourism

**Covers NAICS:** 111219, 111421, 111998, 112910 (apiculture for honey DTC) — usually rolled up to parent farming code

**Modern formats (Part 2):** Christmas Tree Farm, U-Pick Farm / Pumpkin Patch, Maze / Agritourism Farm, Honey / Beekeeper

---

## 1. Data sources (replaces v3 hospitality stack)

| | | | |
|---|---|---|---|
| Google Places / Reviews | Standard hospitality review data | $0.02 | Yelp scrape |
| TripAdvisor | Family-trip reviews + ranking | $0.01 | Claude search |
| State agritourism directory | Named circuits + member benefits | Free | Tourism board |
| Eventbrite / Facebook Events | Local fall / spring event calendar | Free tier | Manual scrape |
| School district directory | Named local schools for field trips | Free | County education dept |
| Wedding venue listing sites | TheKnot, WeddingWire ranking | Free tier | Direct search |
| NOAA frost / pest data | Bloom + harvest window prediction | Free | Extension office |
| Census Bureau county demographics | Family income + young children density | Free | Claude general |

## 2. Competitor radius

30 miles for U-pick / pumpkin patch / Christmas tree (consumers will drive). 60 miles for destination agritourism (corn maze, fall festival, weddings). 100 miles for unique experiences (lavender field, sunflower photography).

## 3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Family / kids friendliness
- Photo quality of experience (Instagrammability)
- Animal interactions
- Parking + access
- Pricing transparency
- Cleanliness of restrooms / facilities
- Wait times for popular activities

- Quality of seasonal product (apples, pumpkins, trees)
- Educational content (school field trips)

## 4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **17** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

| | | |
|---|---|---|
| School field trip program | Named local school districts + curriculum tie-ins | Sauk County School District 4th grade pollinator unit — 12 buses × 50 kids × $8 = $4,800 per Sept-Oct |
| Wedding / event venue licensing | County zoning + state ag tourism laws | Restored barn — 12 weekends × $4,500 = $54k at 0% incremental food cost (caterer brings food) |
| Birthday party packages | Local kids party benchmark prices | $320 package — 10 kids — pony ride + maze + pumpkin — books out 20 weekends |
| CSA / pickup-share | County household demand + delivery zones | 40-share Tuesday pickup CSA — $650/share — $26k from 1 acre — drives traffic to farm stand |
| Influencer hosting program | Named regional Instagram + TikTok creators | Comp 5 trips to @MidwestMomsAdventures (180k followers) — 1 post = ~3,000 farm visits |
| Photography session licensing | Named local photographers + sunflower / fall / Christmas tree photo demand | $50/session photo permit — 60 sessions/season × $50 = $3k zero-effort revenue |
| Value-added product line | On-farm kitchen license + named retail outlets | Apple butter, jam, honey — sold on-farm + at 4 local Whole Foods stores via consignment |
| Goat yoga / farm yoga | Local yoga instructors + Instagram demand | $28/person × 25 people × 18 sessions/yr = $12,600 with farm-side pop-up bar add-on |
| Farm-to-table dinner series | Local chef partnerships + ticket benchmark | $95/person × 40 people × 6 dinners/yr = $22,800 — chef takes 50% of food + 100% of tips |
| Holiday season expansion (Christmas) | Tree species + cut-your-own market | Add wreath-making class + hot cocoa + tractor ride — extends Nov-Dec season + $18 incremental/family |
| Off-season Airbnb / glamping | Yurts / tiny homes / cabin permitting + nightly rate benchmark | 2 yurts × $145/night × 180 nights = $52k off-season revenue |
| Agritourism circuit partnership | Named state circuit + cross-promotion mechanics | WI Cheese / Christmas Tree / Pumpkin trail — listed on state tourism map = +30% traffic during season |
| Mobile e-commerce on-farm | Square / Shopify POS + barcode SKUs | iPad checkout + curbside pickup widget — 22% lift in average ticket vs cash-only |

| | | |
|---|---|---|
| Loyalty / season pass | Local family pass benchmarks (zoo, museum) | $95 family season pass — unlimited fall visits — 320 sold = $30k upfront cash + repeat traffic |
| Festival sponsorship slot | Named regional brands seeking family-event sponsorship | Culver's "Family Saturday" sponsorship — $4k for branded signage + free butter burgers |
| Beekeeping educational program | State 4-H + extension partnerships | $45/person × 12 people × 4 workshops/yr = $2,160 + sells $90/jar honey to attendees |
| Wellness retreat / wedding shower hosting | Named wedding planners + retreat centers | Bridal shower package — $850 for 20 — 18 weekends/yr = $15,300 |

## 5. Specificity rules (L4 — what 'real named entity' means here)

- Every named partner (school, photographer, brand) must be verifiable in Google or state directory
- Every event recommendation must include attendance benchmark from comparable farm in same state
- Every program (CSA, season pass) must include a real benchmark price from an existing operation
- Photography licensing must reference real per-session benchmark in this region

## 6. Cost scale (L5 — per-sector currency bands)

Agritourism cost scale is closer to small retail / hospitality:

| | |
|---|---|
| 1 — Free or under $200 | Sign up for circuit, post on Facebook, comp influencer trip |
| 2 — $200 to $2,000 | Signage, photo permits, simple event hosting, basic marketing |
| 3 — $2,000 to $20,000 | Yurt / tiny home, wedding-prep barn cleanup, POS system |
| 4 — $20,000 to $100,000 | Major facility upgrade, commercial kitchen, parking expansion |
| 5 — Over $100,000 | Wedding barn restoration, lodging buildout, large equipment |

## 7. Persona schema (L6)

**Schema:** B2C_consumer (use the existing v3 hospitality persona schema)

Required persona fields:

- Name + age + family composition
- Distance willing to drive for this experience
- Spend trigger (what makes them open wallet)
- 5-star trigger (what earns the great review)
- Photo behavior (do they share on Instagram / TikTok)
- Word-of-mouth trigger (what makes them tell other parents)
- Lifetime value (visits per year × ticket × food / retail attach)

- Acquisition channel (which named Facebook group / school)

| **Example persona: Megan, 36, Madison mother of 2 kids ages 4 and 7** |
|---|
| Drives 45 min for a "good" fall experience. Spend trigger: pumpkin painting station + free hot cider + Instagrammable hay bale photo wall. 5-star trigger: clean restrooms + short maze line + tractor ride that feels safe. Tells 4-6 other parents in her school WhatsApp group. LTV: 2 visits/yr × $48 (admission + pumpkin + concessions) = $96/yr × 6 yrs while kids are young = $576. |

## 8. Cycle model (L7 — replaces calendar seasons)

Agritourism runs on commodity-product seasonal calendar — different from generic tourism:

| | | |
|---|---|---|
| Pre-bloom (Mar-Apr) | Spring photo bookings + maple syrup | Open early for spring photos, market maple syrup festival |
| Bloom + spring (May-Jun) | School field trip peak | Drive school district sign-ups, run flower-photo days |
| Pick-your-own spring (Jun-Jul) | Strawberry / cherry season | Heavy social media on bloom + harvest dates, partner with bakers |
| Summer dead zone (Jul-Aug) | Heat + competing summer activities | Goat yoga, sunflower fields, farm-to-table dinners — lifestyle pivot |
| Fall peak (Sep-Oct) | Apples, pumpkins, mazes — 60% of annual revenue | Maximize daily ops, weekend volume, school weekday traffic |
| Holiday (Nov-Dec) | Christmas trees + wreaths + cocoa | Cut-your-own + ornament shop + photos with Santa |
| Off-season (Jan-Feb) | Wedding bookings + planning + Airbnb | Wedding venue tours, glamping, planning next year |

## 9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Weather cancellation — universal — refund policy + rain check + insurance
- Parking overflow on peak weekends — universal — early-arrival incentive + shuttle
- Restroom complaints — universal — invest before season
- Animal smell / safety concerns — universal — signage + handwashing stations
- Pricing surprise (gates + activities + photos add up) — frame as bundled pass to remove
- Long lines at popular activities — frame as timed-entry / virtual queue opportunity

## 10. Report sections (L9 — section list and order)

1. **Top 5 unique experiences to add** — Drawn from 17-category playbook
2. **Quick-win operational improvements** — Under $2k actions
3. **Wedding / events licensing roadmap** — If applicable based on county zoning
4. **Agritourism circuit + partnership map** — Named partners within 60 miles

5. **Customer personas (4 personas + lost customer)** — Family / school group / wedding party / commercial buyer
6. **Seasonal cycle plan** — 7-window calendar specific to this farm's product
7. **Known operational challenges** — Weather / parking / restrooms — collapsed drawer

## Key performance indicators worth tracking

- Avg ticket per family (admission + add-ons + food + retail)
- Photo-share rate (Instagram tags per 100 visitors)
- Repeat visit rate within season
- Wedding revenue as % of total (target: 15-30% for ag-tourism + wedding)
- School field trip bookings per fall season
- Off-season revenue % of annual (target: > 10% to smooth cash flow)

# 11.forestry_fishing — Forestry, logging, commercial fishing, support services

**Business model:** B2B_regional | **Profile ID:** agriculture.forestry_fishing

**Covers NAICS:** 113110, 113210, 113310, 114111, 114112, 114119, 114210

**Modern formats (Part 2):** Aquaculture / Fish Farm, Oyster / Shellfish Farm

---

## 1. Data sources (replaces v3 hospitality stack)

| | | | |
|---|---|---|---|
| State DNR forestry program | Named foresters + cost-share programs | Free | Extension office |
| USDA Forest Service | National forest sale schedule + bid history | Free | BLM equivalent |
| NOAA Fisheries / state fish & wildlife | Quota allocation + stock status + permits | Free | Industry assoc. |
| Random Lengths / Madison's Lumber | Pricing for lumber, pulp, chips | Subscription | TSX-listed firms |
| ATFS / FSC / SFI cert bodies | Cert status + premium markets | Free public | Direct cert auditor |
| LandWatch + state timber sale lists | Comparable timberland values + sale prices | Free / Sub | County assessor |
| USDA APHIS | Named pests + quarantine zones (e.g. spongy moth) | Free | State ag dept |

## 2. Competitor radius

Forestry: 100 miles to mill / paper plant. Logging: 60 miles for daily haul. Commercial fishing: defined by quota zone + fishery management area, not geographic radius.

## 3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Stumpage price captured vs regional benchmark
- Cert status (FSC, SFI, ATFS) and price premium realized
- Mill / processor relationships
- Equipment uptime + crew reliability
- Safety record (logging is highest-fatality industry)
- Habitat / regulatory compliance record

## 4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **12** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

| | | |
|---|---|---|
| Named cert pursuit (FSC / SFI / ATFS) | Cost of certification + premium captured in this region | FSC certification — $3,500 audit + $0.50/MBF year-1 — captures premium from named furniture maker |
| Carbon credit registration | Climate Action Reserve / ACR / VCS protocols | CAR Improved Forest Management — $8/ton × 0.5 ton/ac = $4/ac/yr passive income on managed acres |
| Conservation easement income | State + USDA forest legacy programs | Forest Legacy easement — $1,200/ac one-time payment retains ownership + working forest |
| High-value species premium | Named buyers seeking veneer-grade hardwood | Roy Anderson Lumber Wisconsin — pays $1,800/MBF for #1 sugar maple veneer vs $480 for sawlog |
| Logging contract diversification | Mill, pulp plant, biomass facility within 100mi | 4-buyer contract mix (mill + pulp + biomass + firewood) reduces single-buyer concentration risk |
| Recreational lease income | Hunting club + outfitter demand in region | 600-acre hunting lease — $14,000/yr to named local outfitter — zero impact on timber rotation |
| Conservation cost-share enrollment | NRCS EQIP forestry practice list + state programs | EQIP timber stand improvement — $185/ac on 80 ac = $14,800 cost-share for marked thinning |
| Timber stand inventory + management plan | State-licensed forester + cost-share for plan | WI DNR cost-shared forest management plan — $0 to landowner — unlocks $14k of EQIP funds above |
| Niche product (firewood, pellets, mulch) | Local retail demand + bagging / delivery economics | Bagged firewood to 4 named gas stations — $7/bag × 800 bags/mo = $5,600/mo from cull material |
| Custom milling for craftsman market | Wood-Mizer dealer + local custom-furniture market | Custom-mill 80 hardwood logs/yr for named cabinetmakers — $1.20/BF retained vs $0.48 commodity |
| Education / workshop hosting | Land trust + university extension partnerships | Aldo Leopold Foundation forester-day on this property — $1,500 honorarium + reputation lift |
| Storm-damage / salvage rapid response | Named ins. adjusters + emergency services list | Pre-positioned on insurance approved-vendor list — captures storm-salvage premium $/MBF |

## 5. Specificity rules (L4 — what 'real named entity' means here)

- Every named buyer must be a real mill or processor in state forester directory
- Every cost-share program must reference current sign-up window with state forester contact
- Every cert body must be a real ANSI-accredited certifier
- Every species-grade premium must reference current published pricing

## 6. Cost scale (L5 — per-sector currency bands)

Forestry operates on multi-decade cycles. Cost scale is calibrated to a 5-year planning horizon:

| | |
|---|---|
| 1 — Free or under $1,000 | Sign up for program, get cost-shared management plan |
| 2 — $1,000 to $10,000 | Cert audit, inventory work, small equipment |
| 3 — $10,000 to $100,000 | Forwarder / skidder used, sawmill, custom milling shop |
| 4 — $100,000 to $1M | New harvesting equipment, pellet mill startup |
| 5 — Over $1M | Mill expansion, large land acquisition |

## 7. Persona schema (L6)

**Schema:** B2B_buyer (mill procurement, pulp procurement, custom-furniture buyer)

Required persona fields:

- Mill or processor name + procurement officer
- Species + grade demand + annual volume
- Cert preference (FSC / SFI / ATFS)
- Payment terms + scaling method
- Geographic supply shed they pull from
- Long-term agreement appetite

## 8. Cycle model (L7 — replaces calendar seasons)

Forest cycle is dual: short cycle (annual market) + long cycle (rotation length). Both surface in the report:

| | | |
|---|---|---|
| Winter haul (Dec-Mar) | Frozen ground + access to wet sites | Schedule 60% of year's harvest in frozen-ground months |
| Spring breakup (Apr-May) | Mud + access closures | Crew off-time for equipment maintenance + planning |
| Summer (Jun-Aug) | Dry ground + fire season | Ridge / sandy site harvests, fire-management precautions |
| Fall (Sep-Nov) | Pre-freeze prep + planting + cert audits | Plant seedlings, schedule cert recertification |
| 5-yr inventory cycle | Re-cruise and update plan | Full timber inventory + management plan refresh |
| Rotation cycle (40-100 yrs) | Final harvest + regen | Site preparation, replanting, post-harvest residual sale |

## 9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Safety incidents — fatality rate is high — universal
- Weather-driven access loss — universal — frame as winter-haul preparation
- Stumpage price volatility — universal — frame as buyer diversification
- Pest / disease outbreak (spongy moth, EAB, oak wilt) — frame as proactive cert + management
- Labor shortage in skilled forestry crews — universal — frame as crew retention / training
- Insurance availability for logging contractors — frame as safety record / cert as differentiator

### 10. Report sections (L9 — section list and order)

1. **Tract / operation snapshot** — Acres, species mix, current rotation stage, cert status
2. **Top 5 unique opportunities** — Drawn from 12-category playbook
3. **Buyer / mill map** — Named buyers within 100 miles + species fit
4. **Cert + carbon roadmap** — Path to FSC/SFI/ATFS + carbon registration
5. **Cost-share + program inventory** — EQIP, state forestry, Forest Legacy with dollar amounts
6. **Cycle plan** — Annual + rotation calendar with action items
7. **Risk register** — Pest, fire, market, regulatory
8. **Known industry challenges** — Collapsed drawer

### Key performance indicators worth tracking

- Stumpage $/MBF captured vs regional benchmark
- Cert premium realized ($/MBF over commodity)
- Carbon income per acre per year
- Recreational lease income per acre per year
- Days-in-season equipment utilization (target > 180/yr)
- Safety incidents per million hours worked (target = 0)

## Out-of-scope NAICS codes inside this sector

These codes belong to this sector but BizRadar refuses to analyze them — either because they are not real businesses, are regulated monopolies BizRadar cannot help, or do not have customer-facing operations to optimize.

| | |
|---|---|
| 115xxx | Support activities for ag/forestry — analyzed under their parent (e.g. crop harvesting service rolls into agriculture.commodity profile) |

**Engineering note for this sector**

Sector 11 is the first sector where the v3 hospitality pipeline produces obviously broken output. Engineers should expect to spend ~3 weeks on this sector alone — most of the work is sourcing real producer / buyer / program directories rather than writing code. Build the agriculture.agritourism profile first (closest to v3) then commodity, then forestry.

## CHAPTER 21 — MINING, QUARRYING, OIL & GAS EXTRACTION

Industrial extraction operations selling to refiners, smelters, construction. Heavy regulation, capital-intensive, multi-year decision cycles.

*Covers 21 NAICS codes from Part 1 of the master list, plus modern formats from Part 2.*

**The v3 problem in this sector**

A limestone quarry has zero Google reviews and competes within a 200-mile freight radius for asphalt and concrete plants. v3's "competitors within 2.5km" returns nothing useful and the report is empty.

### Sub-clusters in this sector

This sector splits into 2 sub-clusters because the business models inside diverge enough that one profile cannot serve them all. Each sub-cluster gets its own complete spec below.

| | | | |
|---|---|---|---|
| mining.aggregates | Aggregates, sand, gravel, dimension stone, industrial minerals | B2B_regional | 212311-212390 (stone, sand, gravel, clay, kaolin) |
| mining.oil_gas_metals | Oil and gas extraction, coal, metal mining, support services | B2B_national | 211120, 211130, 212114-212290, 213111-213115 |

# 21.aggregates — Aggregates, sand, gravel, dimension stone, industrial minerals

**Business model:** B2B_regional | **Profile ID:** mining.aggregates

**Covers NAICS:** 212311-212390 (stone, sand, gravel, clay, kaolin)

---

#### 1. Data sources (replaces v3 hospitality stack)

| | | | |
|---|---|---|---|
| USGS Mineral Industry Surveys | Production volumes + prices by commodity | Free | State geological survey |
| MSHA Data Retrieval System | Safety record by mine ID + competitor records | Free | OSHA equivalent |
| EPA ECHO | Environmental compliance + violations history | Free | State EPA |
| State permit databases | Reclamation bond + permit status | Free | Direct dept call |

| | | | |
|---|---|---|---|
| NRMCA / NSSGA member directory | Named ready-mix + asphalt buyers | Sub / member | Industry assoc. |
| DOT contract awards | Public-works aggregate contracts | Free | State DOT |
| BlueBook / D&B; | Buyer credit + payment behavior | Subscription | Industry references |
| LandWatch + USGS quad maps | Reserve life + adjacent reserves | Free / Sub | Geologist consult |

## 2. Competitor radius

Aggregates do not freight far. Default radius: 50 miles for ready-mix / asphalt customers; 100 miles for high-spec aggregates (rip-rap, ballast); 200+ miles for high-value industrial minerals (silica, kaolin).

## 3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- On-time delivery rate to ready-mix plants
- Material spec compliance (ASTM C33, AASHTO M80)
- Truck cycle time at scale
- Account payment terms vs competitors
- Safety record (MSHA reportable incidents)
- Reclamation / community standing

## 4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **12** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

| | | |
|---|---|---|
| DOT prequalification + bid pursuit | State DOT certified producer list + naming an account manager | Wisconsin DOT certified aggregate producer — $0 application — qualifies for $80M annual road program bidding |
| Specialty product line addition | Premium spec products (rip-rap, decorative, road base subgrade) | Add limestone rip-rap production — $14/ton vs $7/ton commodity gravel — captures DNR shoreline projects |
| Long-term offtake with ready-mix | Named ready-mix accounts within 50 miles | 5-yr supply agreement with Cemex Madison plant — $2.4M/yr at indexed pricing |
| Reserve expansion / next-pit permitting | Adjacent landowners + state mining permit timeline | Lease 80-acre adjacent reserve from named landowner — $1,200/ac signing + $0.45/ton royalty = 25-yr reserve life extension |
| Reclamation as community-asset narrative | County land trust + post-mining-use plans | Post-mining quarry-as-recreational-lake plan — community PR + state cost-share for closure activities |
| Named ESG reporting + certifications | ISO 14001 + WBA Stone Standard + EPD program | NSSGA Stone Standard Tier-2 certification — qualifies bids on green-building specs (USGBC LEED v4 sourcing credit) |

| | | |
|---|---|---|
| Bagged / DTC consumer channel | Big-box retailer (Home Depot, Menards) supply specs | Add bagged decorative stone — Menards regional supply contract — $4.50/40lb bag retail |
| Recycled aggregate from C&D; | Demolition contractors + state recycled-content specs | Process C&D; concrete — sell at $11/ton vs raw quarried — captures DOT recycled-content credits |
| Trucking-fleet integration | Owner-operator pool + mixed-fleet economics | 4-truck owned fleet — captures $0.18/ton-mile delivery margin on 200k tons/yr = $108k |
| Crushing-on-demand at customer site | Mobile crusher rental + DOT highway-job market | Mobile crushing service to DOT contractors — $2.40/ton vs hauling raw — wins by avoiding haul cost |
| Industrial silica / frac-sand pivot | Sand specs + named oilfield service buyers | High-purity silica for glass — named to Owens-Illinois supply list — $42/ton vs $9/ton construction sand |
| Energy + mine-electrification narrative | Battery-electric haul trucks + state utility programs | Pilot 2 battery-electric haul trucks — XCel Energy rebate + ESG narrative for institutional buyers |

## 5. Specificity rules (L4 — what 'real named entity' means here)

- Every named buyer must be in NRMCA / NAPA / NSSGA member directory or a verified ready-mix / asphalt plant
- Every spec product must reference a real ASTM / AASHTO / state DOT specification number
- Every permit / reclamation reference must name a real state agency contact + permit number range
- Every cert pursuit must name a real auditor + member program

## 6. Cost scale (L5 — per-sector currency bands)

Aggregate operations are heavy-capital. Cost scale reflects industry capex norms:

| | |
|---|---|
| 1 — Under $5,000 | Cert application, DOT prequal, marketing collateral |
| 2 — $5,000 to $50,000 | Lab equipment, scale upgrade, ESG report, marketing video |
| 3 — $50,000 to $500,000 | Mobile crusher, screen plant upgrade, ready-mix lab |
| 4 — $500,000 to $5M | New crusher, conveyor upgrade, reserve land lease |
| 5 — Over $5M | Pit expansion, electrification fleet, new product line buildout |

## 7. Persona schema (L6)

**Schema:** B2B_buyer (procurement officer at named ready-mix / asphalt / concrete-block plant)

Required persona fields:

- Buyer name + role + plant location
- Annual tonnage demand + spec mix
- Decision criteria order (price, on-time, spec compliance, ESG, reliability)
- Pain points with current suppliers
- Contract structure preferred (spot, indexed, fixed)
- Audit / cert requirements

## 8. Cycle model (L7 — replaces calendar seasons)

Aggregate sales mirror construction season + DOT bidding calendar:

| | | |
|---|---|---|
| Winter (Dec-Feb) | Construction off-season + permit / DOT bid prep | Stockpile inventory, file permits, pursue DOT prequal, negotiate offtake |
| Spring (Mar-May) | Construction ramp-up + project starts | Activate spring delivery contracts, surge crew + trucking |
| Summer (Jun-Aug) | Peak construction + DOT season | Maximum production + delivery, safety + heat protocols |
| Fall (Sep-Nov) | Construction wind-down + winter prep | Final-month surge, fall inventory build, equipment maintenance |
| Rolling: DOT bid lettings | State-published bid letting calendar | Submit aggregate bids 6 weeks before letting |
| Annual: permit / reclamation reporting | State filing deadlines | Reclamation progress + bond review |

## 9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Community / NIMBY pushback — universal — frame as proactive community relations
- Permit + reclamation cost — universal — frame as planning + bond reduction strategies
- Truck driver shortage — universal — frame as driver retention + autonomous haul pilot
- Diesel cost volatility — universal — frame as fuel surcharge + electrification
- Reserve depletion — universal but timeline-specific — frame as adjacent leases
- Mine safety record visibility — universal — frame as MSHA-zero campaign for procurement

## 10. Report sections (L9 — section list and order)

1. **Operation snapshot** — Reserves remaining, products, named buyers, tonnage trend
2. **Top 5 differentiation moves** — Drawn from 12-category playbook
3. **Buyer + DOT pipeline** — Named accounts + DOT bid letting calendar
4. **ESG / cert roadmap** — Named cert + premium realization plan
5. **Permit + reclamation calendar** — Filing deadlines + bond optimization
6. **Fleet / equipment plan** — Replacement timing + electrification pilots
7. **Safety scorecard** — MSHA + lost-time-incident benchmark
8. **Known industry challenges** — Collapsed drawer

### Key performance indicators worth tracking

- Tons sold per year + average $/ton realized
- On-time delivery rate to top 5 customers (target > 97%)
- MSHA reportable incidents per 200k hours (target = 0)
- DOT prequalification status + active bids count
- Reserve life remaining (years at current production)
- ESG cert coverage % of production (target > 50%)

# 21.oil_gas_metals — Oil and gas extraction, coal, metal mining, support services

**Business model:** B2B_national | **Profile ID:** mining.oil_gas_metals

**Covers NAICS:** 211120, 211130, 212114-212290, 213111-213115

## 1. Data sources (replaces v3 hospitality stack)

| | | | |
|---|---|---|---|
| EIA (Energy Information Admin) | Production + price data by basin / county | Free | State oil + gas comm |
| USGS Mineral Industry Surveys | Metal production + reserves | Free | State geological survey |
| MSHA Data Retrieval | Safety + quarterly reports | Free | OSHA |
| EPA ECHO + state DEQ | Environmental compliance | Free | State EPA |
| SEC EDGAR (for public co) | Reserves + production reports | Free | 10-K of named operator |
| London Metal Exchange / NYMEX / CME | Commodity pricing + futures | Sub / public | Vendor screen |
| BLM / state agency lease databases | Lease status + bonus + royalty | Free | BLM regional office |
| Drilling Info / Enverus | Wells, leases, production by operator | Subscription | State commission |

## 2. Competitor radius

Oil + gas: basin-defined (Permian, Marcellus, Bakken, etc.). Metals: national or global by commodity. Coal: regional (Powder River, Illinois, Appalachian).

## 3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- Operating cost per BOE / ton produced
- Reserves replacement ratio
- Safety + environmental compliance record
- Joint-venture / partner reputation
- Regulator + community standing
- ESG rating from named agencies (Sustainalytics, MSCI)

## 4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **12** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

| | | |
|---|---|---|
| ESG / sustainability disclosure upgrade | TCFD + SASB + GRI standards adoption | Adopt SASB Mining standard disclosures — captures institutional investor flow + green bond eligibility |
| Methane abatement narrative (oil + gas) | EPA OOOOb + state methane rules + EU CBAM exposure | OOOOb-compliant tanks + LDAR — captures premium $/Mcf from named LNG buyer requiring low-methane gas |
| Long-term offtake / royalty trust | Named refinery + smelter buyers + capital-markets structures | 10-yr take-or-pay with named refinery — locks margin + qualifies for project finance |
| Carbon-credit registration (CCS, CCUS) | 45Q tax credit + named CCS partners | 45Q $85/ton sequestered — co-develop with named midstream operator — $24M/yr at 280k tons |
| Tribal / community agreements (FPIC) | Named tribal nations + ICMM principles | Free Prior Informed Consent agreement with named tribal council — unlocks $40M federal grant for project |
| Battery-minerals positioning (Li, Co, Ni, Cu, REE) | IRA Section 30D battery sourcing + named OEM offtake | IRA-qualifying lithium spodumene supply to GM Ultium battery JV — $2,600/ton vs $1,400 spot |
| Critical-minerals federal program enrollment | DPA Title III + DOE LPO + DoD Industrial Base Fund | DPA Title III $30M cost-share for high-grade graphite refining |
| Reclamation bond optimization | Self-bonding + corporate guarantee alternatives | Replace cash bond with corporate surety — frees $14M working capital at 1.5% premium cost |
| Mineral lease consolidation | Adjacent operator landman + landowners + JV structures | Consolidate fragmented leases via 2-mile lateral pooling — increases recoverable reserves 18% |
| Royalty-and-streaming financing | Wheaton / Franco / Royal Gold / private royalty buyers | $60M streaming deal with named royalty company — non-dilutive growth capital at 5-7% cost |
| Tour + community education program | County school district + state museum partnerships | Quarterly mine tour + scholarship program — addresses local opposition + builds talent pipeline |
| Tailings repurposing / secondary recovery | Tailings reprocessing tech + secondary metal markets | Reprocess 800-acre tailings pile for residual copper — $4M capex + $1.8M/yr revenue + reclamation credit |

## 5. Specificity rules (L4 — what 'real named entity' means here)

- Every named buyer must be a real refinery / smelter / OEM / utility
- Every cert / standard must reference real body (SASB, ISO 14001, IRMA)
- Every program must reference real federal agency and program name
- Every cost / royalty figure must reference public benchmark (EIA, USGS, S&P;)
- Every tribal / community agreement must reference real nation and contact

## 6. Cost scale (L5 — per-sector currency bands)

Oil/gas/metals operate at industrial capex scale. Bands are calibrated accordingly:

| | |
|---|---|
| 1 — Under $50,000 | Cert application, ESG report, community PR |
| 2 — $50,000 to $500,000 | Methane LDAR, stakeholder engagement, study |
| 3 — $500,000 to $5M | Equipment retrofit, expanded compliance, pilot project |
| 4 — $5M to $50M | Major equipment, capacity expansion, JV stake |
| 5 — Over $50M | New mine, refinery, processing facility |

## 7. Persona schema (L6)

**Schema:** B2B_buyer (refinery, smelter, OEM procurement, utility)

Required persona fields:

- Buyer org + named procurement / commercial contact
- Volume needed + spec window
- Cert / ESG requirements
- Term length preference + renewal options
- Pain points with current supply portfolio
- Strategic priorities (security of supply, ESG, cost)

## 8. Cycle model (L7 — replaces calendar seasons)

Industrial extraction runs on commodity-price + permit + capital-allocation cycles, not seasons:

| | | |
|---|---|---|
| Annual: AFE + capital plan | Board / partner capital allocation | Submit AFEs by Q4 for next year capital |
| Annual: regulator + bond review | State + federal annual filings | File reclamation + ops reports |
| Quarterly: SEC / SEDAR (public) | Reserve report + production update | Coordinate IR + analysts |
| Multi-year: permit + EIS cycle | 3-7 year EIS / permit timeline | Stakeholder engagement + biological surveys |
| Commodity cycle (3-10 yr) | Macro price cycle peaks + troughs | Counter-cyclical capex, hedging program |
| Election cycle | Federal + state regulatory shifts | Trade group engagement, scenario planning |

## 9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Commodity price exposure — universal — frame as hedging + diversification

- Community + tribal opposition — universal — frame as proactive consultation
- Permit + EIS delay — universal — frame as parallel-pathing + contingency
- Reclamation cost overrun — universal — frame as bond optimization
- Worker safety + fatality risk — universal — frame as zero-harm program
- Capital availability swings — universal — frame as alternative financing
- Methane / carbon scrutiny — emerging — frame as abatement narrative

## 10. Report sections (L9 — section list and order)

1. **Asset snapshot** — Reserves, production, named buyers, ESG profile
2. **Top 5 strategic moves** — Drawn from 12-category playbook
3. **Buyer + offtake map** — Named buyers + contract status + premium opportunities
4. **ESG + cert roadmap** — Path to recognized ESG ratings + premium markets
5. **Permit + community engagement plan** — Stakeholder map + filing calendar
6. **Capital structure + financing options** — Streaming / royalty / project finance / federal cost-share
7. **Risk register** — Price, permit, community, safety, reclamation
8. **Known industry challenges** — Collapsed drawer

## Key performance indicators worth tracking

- Operating cost per BOE / ton produced vs basin / commodity benchmark
- Reserves replacement ratio (added ÷ produced, target > 1.0)
- Total recordable incident rate (target = 0)
- ESG rating from named agency (target: top quartile)
- Methane intensity (kg $CO_2e$/Mcf for gas, target < 0.20)
- Community-grievance log open items (target = 0)

| Engineering note for this sector |
| --- |
| Sector 21 is rarely a v3 customer because mining + oil/gas operations have full-time strategy teams. The v4 value proposition is for small / mid-cap operators (single-mine, single-basin) who lack that strategy depth. Phase 5 work — build last unless a specific request arrives. |

## CHAPTER 22 — UTILITIES (ELECTRIC, GAS, WATER, SEWAGE)

Regulated monopolies and utility cooperatives. The "competition" framing is regulatory + customer-satisfaction, not market-share.

*Covers 14 NAICS codes from Part 1 of the master list, plus modern formats from Part 2.*

**The v3 problem in this sector**

A regulated electric utility has a 100% market share in its franchise area. v3's "competitor gap analysis" is meaningless. v4 reframes the analysis as regulator + customer + grid-modernization optimization.

# 22.electric — Electric utilities (IOU, coop, municipal) and gas / water utilities

**Business model:** Regulated_monopoly | **Profile ID:** utilities.electric

**Covers NAICS:** 221111-221122, 221210, 221310-221330

### 1. Data sources (replaces v3 hospitality stack)

| | | | |
|---|---|---|---|
| FERC eLibrary + Form 1 | Rates, transmission, financial | Free | State PUC |
| EIA Form 861 / 923 | Sales, generation, customer counts | Free | Annual report |
| State PUC dockets | Rate cases + complaints + IRP filings | Free | Direct PUC office |
| J.D. Power utility studies | Customer satisfaction benchmarks | Subscription | ACSI alternative |
| SAIDI / SAIFI public data | Reliability metrics + outage history | Free EIA / state | Internal SCADA |
| NERC reliability standards | Compliance + incident history | Free public | Audit report |
| NRECA / APPA / EEI | Industry benchmarks by class | Member-only | Public summary |
| DOE LBNL utility cost data | O&M; and capital benchmarks | Free public | EIA |

### 2. Competitor radius

Same NERC region for reliability benchmarking. Same state for rate-case comparison. Same utility class (IOU vs coop vs muni) for governance benchmarking.

### 3. Review category taxonomy (L2A tags)

Reviews are tagged into these categories instead of v3's hotel-shaped set:

- SAIDI / SAIFI vs region
- Rate level vs region
- Customer service responsiveness
- Billing accuracy + clarity
- Outage communication during emergencies
- Program participation rates (DR, EE, low-income)
- Capital investment efficiency

## 4. Opportunity playbook (L3 — replaces the 18 hospitality categories)

This sector researches **14** opportunity categories. Every report must produce at least 12 specific ideas drawn from at least 8 of these categories.

| | | |
|---|---|---|
| Time-of-use + critical-peak rate adoption | Named regulatory model + customer adoption rates | TOU rate offering — modeled on PG&E; TOU-D-PRIME — 28% bill savings for EV-owning customers |
| EV charging network build-out | NEVI funds + utility-owned charging program | NEVI $5M corridor charging award + $0.18/kWh DCFC tariff revenue |
| Distributed generation / community solar | Named developers + state community-solar program | Community solar program — 20MW in three sites — adds $2.4M annual margin + low-income subscription |
| Demand response + virtual power plant | Named DR aggregators (CPower, Voltus, OhmConnect) | 40MW residential DR via OhmConnect — $80/kW-yr capacity payment + grid services revenue |
| AMI / smart meter deployment | AMI vendor mix + business case | AMI 2.0 deployment — Itron Riva — $80M capex + $14M annual O&M; savings + customer engagement features |
| Grid resilience / storm hardening | FEMA + state hardening grants + IIJA funds | IIJA Grid Resilience Innovation Partnerships $20M award — undergrounding 60mi feeder + microgrid |
| Battery storage as transmission asset | FERC Order 841 + ISO market participation | 40MW/160MWh battery as T&D; deferral — 4-yr defers $30M substation upgrade + $4M/yr energy arbitrage |
| Microgrid offerings to commercial | Named anchor customer (hospital, military, university) | Hospital microgrid pilot — $14M capex + $2.4M annual reliability premium revenue |
| Energy efficiency program expansion | State EE goal + named program portfolio | Expand HVAC tune-up rebate — 12k participants/yr — meets state EE goal + earns performance incentive |
| Low-income customer programs | LIHEAP partnership + state weatherization | CARE program partnership with state LIHEAP — 22k enrolled — bills 25% lower + reduces uncollectibles 18% |

| | | |
|---|---|---|
| Dark-fiber leasing | Existing fiber inventory + telecom market | Lease excess dark fiber to Verizon + 3 ISPs — $1.8M/yr non-rate revenue |
| Heat pump electrification campaign | Named HVAC contractor network + state rebate | Heat pump promo — partner with 14 named HVAC contractors — $1,800 utility rebate + $2k state — 4,000 installs/yr |
| Wildfire mitigation + PSPS protocol (West) | CalFIRE + state utility commission rules | Multi-year wildfire mitigation plan + PSPS protocol — recovers $400M/yr in rate base |
| Rate-case PR + stakeholder engagement | Named consumer advocates + intervenor groups | Pre-filing stakeholder workshops with named advocate orgs — improves rate-case approved % by ~12 points |

## 5. Specificity rules (L4 — what 'real named entity' means here)

- Every program must reference real state PUC docket + named regulator division
- Every grant / fund must reference real federal program (NEVI, IIJA, DOE LPO, IRA 48E)
- Every named partner / aggregator / vendor must be a real company
- Every benchmark must reference EIA, J.D. Power, EEI, or NRECA published data
- Every named ratepayer / advocate must be real intervenor in this PUC docket history

## 6. Cost scale (L5 — per-sector currency bands)

Utilities operate at infrastructure scale. Cost bands reflect typical capex programs:

| | |
|---|---|
| 1 — Under $100,000 | Customer comm program, study, regulator filing prep |
| 2 — $100,000 to $1M | Pilot DR, small EE program, dark-fiber buildout |
| 3 — $1M to $10M | Microgrid pilot, AMI software, grid-edge platform |
| 4 — $10M to $100M | AMI deployment, substation rebuild, microgrid full |
| 5 — Over $100M | Generation asset, major T&D; rebuild, AMI 2.0 system-wide |

## 7. Persona schema (L6)

**Schema:** B2B_buyer (regulator + large customer) plus B2C_consumer (residential ratepayer)

Required persona fields:

- Stakeholder type (PUC commissioner, large industrial, residential, advocate, intervenor)
- Decision authority + influence
- Top concern in their constituency / portfolio
- Past dockets / positions taken
- Engagement preference (formal filing, workshop, op-ed, direct meeting)
- Risk to utility if mishandled

## 8. Cycle model (L7 — replaces calendar seasons)

Utilities run on regulatory + capital + storm cycles, not calendar:

| | | |
|---|---|---|
| Rate-case cycle (3-5 yrs) | PUC-defined window | Pre-file workshops, file rate case, intervene, rebuttal, brief, order |
| Annual: IRP filing | State integrated resource plan | Long-term resource portfolio + clean-energy transition |
| Annual: capital plan | Board capital approval | AFE list + ROE deployment + grant pursuits |
| Storm season (varies by region) | Hurricane / tornado / wildfire | Pre-position crews, emergency comm protocol, mutual aid |
| Peak demand season | Summer (Sun Belt) / winter (Northeast) | DR activation, mutual aid agreements, operating reserves |
| Election cycle | State / federal regulatory shifts | Industry advocacy + commissioner education |

## 9. Issue library (L8 — known category-standard complaints)

These complaints are universal to this sub-cluster. They appear in the bottom drawer of the report — never in the lead.

- Reliability outages during storms — universal — frame as resilience + restoration
- Bill spikes during weather extremes — universal — frame as TOU + EE + bill protection
- Rate increases / affordability — universal — frame as program help + transparency
- Customer service hold times — universal — frame as digital self-service + AMI
- Aging infrastructure — universal — frame as grid-modernization narrative
- Wildfire liability (Western states) — universal — frame as PSPS + mitigation
- Clean-energy transition pace — varies — frame as pragmatic IRP + workforce

## 10. Report sections (L9 — section list and order)

1. **Utility snapshot** — Customer count, peak load, generation mix, recent rate-case outcome
2. **Top 5 strategic moves** — Drawn from 14-category playbook
3. **Stakeholder map** — PUC commissioners, advocates, large customers, intervenors
4. **Federal funding pipeline** — IIJA / IRA / DOE programs with submission calendar
5. **Reliability + storm response plan** — SAIDI / SAIFI targets + restoration playbook
6. **Customer program portfolio** — EE / DR / EV / community solar + benchmarks
7. **Rate-case roadmap** — Pre-filing, intervenor management, key witness list
8. **Known industry challenges** — Collapsed drawer

## Key performance indicators worth tracking

- SAIDI minutes (target: top quartile in NERC region)
- SAIFI events (target: top quartile in NERC region)
- J.D. Power customer satisfaction index (target: top tertile)
- EE program savings (% of sales, target meets state goal)
- Federal funding awarded (NEVI / IIJA / IRA)
- Rate-case ROE approved vs requested (% of ask)

## Out-of-scope NAICS codes inside this sector

These codes belong to this sector but BizRadar refuses to analyze them — either because they are not real businesses, are regulated monopolies BizRadar cannot help, or do not have customer-facing operations to optimize.

| | |
|---|---|
| 221121 | Bulk power transmission — typically operated by ISO/RTO, not a customer-facing business BizRadar can help |

**Engineering note for this sector**

Sector 22 is the smallest customer pool (only ~3,000 utilities in the US) but they are extremely high-LTV. The v4 value prop is for small/mid coops + munis (EnergyTrust-tier customers). Phase 5 work — pursue only after a co-op trade association expresses interest.